

LAURA BURNETT

Research Data Scientist | Research Software Engineer

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PROFILE

I am a research data scientist and software engineer with experience of managing biological data. I love sifting through complex datasets to uncover novel insights, and building software that makes data usable and informative for a broader audience. I care deeply about the FAIR principles, and I am adept at rapidly navigating new scientific domains. I want to join a dynamic, collaborative team where I can extend my technical skills and build open-source tools with real-world impact.

SKILLS

Languages

Python (pandas, NumPy, Plotly) • MATLAB • SQL • Cypher (Neo4j) • JavaScript, HTML & CSS

Software practice

Git & pull-request workflow • continuous integration (GitHub Actions) • packaging & dependency management (pyproject, pixi) • technical writing & documentation

Data & methods

ETL & pipeline design • statistical modelling • time-series & signal processing • image analysis • deep learning for automated annotation (DeepLabCut) • dimensionality reduction & clustering

Visualisation

Interactive web resources • dashboards • SVG & JavaScript visualisation • publication quality figures

Core competencies

Reproducible & executable analyses • FAIR data principles • open-source development • mentoring & teaching • communicating technical work to non-specialists

RESEARCH & ENGINEERING EXPERIENCE

Postdoctoral Research Scientist

Reiser Lab, HHMI
Janelia Research
Campus

📅 Aug 2023 – May 2026

📍 Ashburn, VA, USA

- Connectomics & graph databases
- *Drosophila* visual neuroscience
- High-throughput behavioural screening
- Interactive web resources
- Reproducible workflows

- **Built *neuView*, an open, web-based research tool.** An open source Python application that turns a Neo4j connectome database into a browsable page for every cell type in the fly nervous system, containing pre-computed statistics, interactive graphics and connectivity tables. Selected contributions include the embedded Neuroglancer 3D view with interactive features and the hexagonal occupancy heatmaps extended with JavaScript so they can be explored layer by layer. Tested with pytest, built and deployed through GitHub Actions.
- **Developed code to analyse a *Drosophila* connectome — classifying ~53,000 neurons into 732 cell types.** One of a core team of 12. Specifically contributed the spatial patterning analysis to understand how every cell type samples the visual field. Published in *Nature* (2025) as an *executable paper*, with every figure reproducible from a public repository. My plotting code was later adopted by an external connectome-analysis package, *connectome_interpreter*.
- **Designed and ran a high-throughput behavioural screen.** Built the entire pipeline: stimuli for the lab's custom LED arenas, acquisition scripts, and synchronisation across camera, stimulus and environmental logging, through to QC, statistical analysis and figure generation. Added dashboards for browsing the data and wrote the public documentation. The rig and pipeline are now in use by new members of the group running their own experiments, and an accompanying paper is in preparation.
- **Wrote research software for a patch-clamp electrophysiology collaboration.** A two-pass adaptive protocol: the first pass maps a cell's receptive field, the second generates stimuli in real time, confined to that region. The documentation I created for this project became a crucial shared resource among the various team members.
- **Tested new open-hardware LED displays for visual neuroscience.** Joined a small team developing the sixth generation of the lab's modular LED arena as its experimental tester, writing the code that verified new panels could render and run visual stimulus patterns.

PhD Researcher

Joesch Lab, Institute of Science and Technology Austria (ISTA)

📅 Sep 2016 – May 2023

📍 Klosterneuburg, Austria

- Neurodevelopment, visual perception & decision-making
- *In vivo* electrophysiology
- Optogenetics
- Computational behavioural analysis

- **Led a multi-year study from question to publication.** Asked whether genetically distinct mouse models of autism share the same perceptual impairments, and where in the brain they arise. Owned the experimental design and analysis plan, and formed a new collaboration between two research groups to deliver it. First author in *PLoS Biology* (2024). I also built the experimental rigs I worked on.
- **Trained and validated deep-learning pose-estimation models** (*DeepLabCut*) to track pupil position and full-body pose in behaving mice. Designed the annotation strategy, tested how well models generalised across sessions and animals, and built the pipeline converting raw tracking into quantitative behavioural measures.
- **Built analysis pipelines for large, noisy multimodal data** — video, behavioural and electrophysiological time series — from raw signals through to statistical inference. Wrote all of my own analysis code, and developed guidelines and best practices that were adopted by incoming members of the lab.
- **Deliberate breadth:** rotations in bulk RNA-seq analysis, microfluidic device fabrication and two-photon microscope assembly, alongside advanced statistics coursework during my first year.

Earlier research positions

UCL Institute of Child Health · University of Melbourne · The Florey Institute

📅 2014 – 2016

📍 London, UK & Melbourne, Australia

- **Projects related to nervous system development, enteric and cortical, with quantitative analysis of confocal microscopy image data.** Four research placements across three institutions — Master's project with the Burns Lab (UCL ICH), undergraduate placements with the Young Lab (Melbourne) and Tan Lab (Florey), and a graduate summer internship with the Hippenmeyer Lab (ISTA) — contributing to publications in *Neuron*, *Nature Communications* and *Developmental Biology*.

SOFTWARE & OPEN SOURCE

- [neuView](#) — Python application generating a static, interactive cell-type explorer for a Neo4j connectome database of >150,000 cells, with a pytest suite, GitHub Actions CI, packaged, GPL-3.0. Deployed at reiserlab.github.io/celltype-explorer-drosophila-male-cns.
- [reiser-documentation](#) — Documentation site covering experimental protocols, analysis code and rationale for a research lab; written as a reference and for people joining the work.

TEACHING & MENTORING

- **Supervision and mentoring:** twelve people across two institutions — one research technician, one graduate student, six graduate rotation students, two summer undergraduates and two high-school summer students (Janelia 2023–2026; ISTA 2016–2023).
- **Teaching Assistant**, Neuroscience Track Core Course, ISTA (2018–2020): taught the image analysis with MATLAB section to incoming first-year graduate students.
- **Summer camp session lead**, ISTA Fakebusters (2021, 2022): designed and led a one-day session, “*Seeing is believing? Vision and optical illusions*”, for ~25 high-school students.
- **Mentoring group member**, Janelia Women and Non-binary Mentoring Group (2023–present, now remote alumni member).

TALKS & PRESENTATIONS

Invited talks: joint symposium with the Champalimaud Centre for the Unknown (2021); joint symposium with Universität für Bodenkultur Wien (BOKU), ISTA (2021).

Posters: NeuroFly, Birmingham (2024); FENS Forum, Paris (2022) and online (2020); FENS International Winter School, Sölden (2022); Austrian Neuroscience Meeting, Salzburg (2021) and Innsbruck (2019); Colour Vision: Circuits and Behaviour, HHMI Janelia (2019).

Internal seminars and lab meetings, ISTA and HHMI Janelia (2016–2026).

SELECTED PUBLICATIONS

Full list: [Google Scholar](#) · h-index 5 · 408 citations.

1. Berg S, Beckett IR, Costa M, ... **Burnett LE**, et al. Sexual dimorphism in the complete *Drosophila* male central nervous system connectome. *Cell*. 2026;189(18):5504–5526.
2. Nern A, Loesche F, Takemura S, **Burnett LE**, et al. Connectome-driven neural inventory of a complete visual system. *Nature*. 2025;641(8065):1225–1237.
3. **Burnett LE**, Koppensteiner P, Symonova O, et al. Shared behavioural impairments in visual perception and place avoidance across different autism models. *PLoS Biology*. 2024;22(6):e3002668.

EDUCATION

PhD, Neuroscience

[Institute of Science and Technology Austria \(ISTA\)](#) ☞ Sep 2016 – May 2023 📍 Klosterneuburg, Austria

Thesis: *To flee, or not to flee? Using innate defensive behaviours to investigate rapid perceptual decision-making through subcortical circuits in mouse models of autism.*

MSci Natural Sciences — Neuroscience (First Class Honours)

[University College London \(UCL\)](#) ☞ Sep 2015 – May 2016 📍 London, UK

BSc Natural Sciences — Organic Chemistry & Biomedical Sciences (First Class Honours)

[University College London \(UCL\)](#) ☞ Sep 2012 – May 2015 📍 London, UK

[University of Melbourne \(study abroad, Dean's List\)](#) ☞ July 2014 – July 2015 📍 Melbourne, Australia

A-Levels — Mathematics (A*), Biology (A*), Spanish (A*), Chemistry (A)

[King's High School, Warwick](#) ☞ 2010 – 2012 📍 Warwickshire, UK

OUTREACH & COMMUNITY

- **Volunteer maths tutor**, Action Tutoring (2026–present): weekly Year 6 maths tuition for students from disadvantaged socio-economic backgrounds.
- **STEM outreach**, HHMI Janelia (2024–2026): “Hour of Code”, teaching primary-school children to code in Python; RESET / ScienceExplorers, hands-on inquiry-based STEM activities for 5th-grade students.
- **Science communicator**: founding member of *Wissen Schafft's* (2021), a multilingual YouTube series created with ISTA colleagues to tackle misinformation during the COVID-19 pandemic.
- **Founder and organiser**, ISTA Art Club (2018–2023); **Graduate Student Association Representative**, ISTA (2018–2020).
- **Event volunteer**, EX2 Adventures (2024–2026): trail running and mountain bike races in Virginia and Maryland.

FUNDING & AWARDS

Travel grant, €500, Austrian Neuroscience Association, FENS (2022).

Austrian Agency for Education and Internationalisation (OeAD) research scholarship €3000 — competitive, self-authored proposal (2016).

Dean's List, University of Melbourne (2014) — awarded for outstanding academic achievement, placing in the top 3–5% of the cohort.